
NASH BARGAINING PREFERENCE OPTIMIZATION

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ABSTRACT

Aligning large language models (LLMs) to multiple objectives requires both representing preferences within each objective and selecting a compromise across objectives. Human comparisons need not admit a global scalar reward: annotators may weigh latent attributes differently across contexts and response pairs, so aggregate judgments can be cyclic. Most existing methods nevertheless represent each objective through scalar scores over individual responses and cannot capture such cycles. Nash Learning from Human Feedback (NLHF) instead treats prompt-conditioned pairwise preferences directly as game payoffs. A recent multi-objective NLHF approach retains this generality but uses max-min aggregation. However, max-min can select Pareto-dominated outcomes and change under objective rescaling, so the resulting compromise lacks basic efficiency and scale robustness. We introduce Nash Bargaining Preference Optimization (NBPO), which separates these decisions. An entropy-regularized preference game yields one value per objective, and Nash bargaining balances improvements over a reference fallback. The reference becomes a meaningful status quo, while the trade-off follows proportional gains rather than prespecified weights. When joint improvement is feasible, NBPO strictly improves every objective over the fallback, selects a Pareto-optimal and unique game-value vector, is proportionally fair, and leaves the set of optimal policies unchanged under independent positive affine transformations of objective utilities. A closed-form regularized opponent avoids training an additional opponent model, while partition-free pairwise log-ratio regression removes policy normalizers and learns from binary comparisons without fitting scalar rewards. The resulting constrained entropy-proximal solver starts directly from the reference, recovers inverse-surplus weights, and admits last-iterate convergence with explicit sampling and regression errors.

1 INTRODUCTION

Preference-based post-training is central to aligning Large Language Models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard Reinforcement Learning from Human Feedback (RLHF) learns a scalar reward from pairwise comparisons, while Direct Preference Optimization removes the separately trained reward model but retains an implicit scalar reward and a Bradley–Terry (BT) representation of preferences (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches require comparisons to be explained by a single scalar score assigned to each response. This assumption is restrictive for multi-objective alignment, where one deployed assistant must satisfy objectives such as helpfulness, harmlessness, correctness, and conciseness.

Even within one named objective, human judgments may depend on several latent attributes whose relative importance changes across annotators, prompts, and response pairs. Consider three answers to the same prompt under a helpfulness criterion. Response A is direct but omits important qualifications, response B is comprehensive but verbose, and response C is clear and well qualified but less complete. Aggregate comparisons may prefer A to B for relevance, B to C for completeness, and C to A for clarity and reliability. Such a cycle can reflect genuine pair-dependent trade-offs rather than disposable annotation noise. No scalar ranking can preserve all three relations: any BT reward model must reverse or tie at least one edge and thereby project the feedback onto a transitive ordering. Optimizing this projection aligns the policy to the fitted ordering rather than the original comparison structure, so a scalar-optimal response can remain vulnerable to another response under the observed preferences.

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, while INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a separate question: after representing each objective faithfully, which compromise should one deployed policy realize? Existing scalar multi-objective methods require an external trade-off specification, such as weights, priorities, or constraints (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026), whereas recent game-theoretic extensions use Blackwell target sets or max-min aggregation (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026). Fixed-reference methods such as MOPO are computationally convenient but can discard pairwise structure that is invisible from the reference (Agnihotri et al., 2026; Zhang et al., 2026).

We study the complementary setting in which a fixed objective panel must be combined into one policy without prespecified priorities. Here, representing preferences within each objective and aggregating values across objectives are distinct decisions. The first calls for a game value that retains the full pairwise structure; the second calls for an explicit compromise rule. A max-min game couples the two by imposing a weakest-objective rule while evaluating the policy. It can retain Pareto-dominated tied solutions and can change under independent rescaling of objective utilities even when the underlying preferences are unchanged.

We introduce Nash Bargaining Preference Optimization (NBPO), which separates these roles. For each objective, an adaptive opponent searches for responses that expose weaknesses of the current policy, while entropy regularization keeps the opponent near the deployed reference. This produces a game value based on the full pairwise comparison matrix rather than a scalar response score or only a fixed-reference margin. The reference also defines the bargaining fallback, and NBPO selects a policy by balancing positive game-value improvements over it. Nash bargaining is natural because it evaluates each objective relative to a meaningful status quo, gives more influence to objectives with smaller gains, and requires no externally chosen objective weights.

When joint improvement over the reference is feasible, every NBPO optimizer improves every objective strictly beyond the fallback and is Pareto optimal in game-value space. Different policies may realize this outcome, but the selected game-value vector is unique. The policy selection is proportionally fair and unchanged by independent positive affine transformations of the objective utility scales. Utilitarian and max-min aggregation are useful alternatives, but neither guarantees this full combination of fallback-relative protection, efficiency, uniqueness, and scale covariance without additional restrictions or tie-breaking.

The optimizer preserves the same separation. We impose positive game-value improvement as a constraint and solve entropy-proximal bargaining subproblems directly from the reference boundary, rather than optimizing a separate warm-start objective. Under joint improvement, the first constrained subproblem enters the positive-surplus region. Entropy regularization yields a closed-form opponent, and the KKT multipliers recover inverse-surplus objective weights. A low-dimensional dual update and partition-free pairwise log-ratio regression then implement each policy step from binary comparisons without fitting scalar rewards. Simultaneous confidence intervals certify accepted neural stages, and the convergence analysis covers exact last iterates as well as approximate updates with explicit sampling and regression errors.

Matched finite-temperature runs on HELPSTEER2 and SAFERLHF instantiate the pipeline and diagnose how the endogenous bargaining weights respond to objective conflict. The reported runs achieve positive objective improvements but do not establish uniform dominance over fixed aggregation rules, so we treat them as implementation and diagnostic evidence rather than a blanket superiority claim. Overall, NBPO provides an end-to-end separation between objective-wise games that preserve potentially cyclic feedback and a reference-relative bargaining rule that selects, optimizes, and certifies one multi-objective policy.

2 BACKGROUND

General pairwise preference oracles. A prompt x is drawn from ρ , and a policy $\pi(\cdot \mid x)$ is a distribution over a finite response set $\mathcal{Y}_x$. For the finite-dimensional population analysis, we take

the support of ρ and every $\mathcal{Y}_x$ to be finite; the sampled implementation replaces these population expectations by empirical ones. Let $\Pi = \prod_{x \in \text{supp}(\rho)} \Delta(\mathcal{Y}_x)$ denote the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference oracle $P_k(y \succ z \mid x) \in [0, 1]$ satisfying

$$P_k(y \succ z \mid x) + P_k(z \succ y \mid x) = 1, \quad P_k(y \succ y \mid x) = \frac{1}{2}. \quad (1)$$

Define the centered payoff

$$A_k(y, z \mid x) = P_k(y \succ z \mid x) - \frac{1}{2}, \quad A_k(y, z \mid x) = -A_k(z, y \mid x), \quad (2)$$

and, for policies π and ν ,

$$g_k(\pi, \nu) = \mathbb{E}_{x \sim \rho, y \sim \pi(\cdot \mid x), z \sim \nu(\cdot \mid x)} A_k(y, z \mid x). \quad (3)$$

No scalar reward representation or transitivity assumption is imposed. We write the contextual policy inner product as $\langle q, \pi \rangle_\rho = \mathbb{E}_x \sum_y \pi(y \mid x) q(y \mid x)$ and the contextual KL analogously.

Bargaining problems. A K -dimensional bargaining problem is a feasible utility set $\mathcal{G} \subseteq \mathbb{R}^K$ and disagreement point $d \in \mathcal{G}$. It is nondegenerate if some $u \in \mathcal{G}$ satisfies $u > d$ componentwise. On a compact, convex, comprehensive problem, the Nash solution maximizes

$$\sum_{k=1}^K \log(u_k - d_k) \quad (4)$$

over utilities strictly above disagreement (Nash, 1950). Its derivative in coordinate k is the inverse surplus $1/(u_k - d_k)$.

3 OBJECTIVE-WISE PREFERENCE GAMES

This section turns each objective-wise preference oracle into a scalar policy utility by defining an entropy-regularized preference game, deriving its adaptive opponent and policy gradient, and illustrating why this utility captures cyclic structure that fixed-reference evaluation can miss. The reference policy μ has two distinct roles. It is the deployed fallback used by the bargaining problem, and it anchors the opponent distribution used to evaluate each objective. Sharing the same policy does not merge these roles.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{g_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \rho} \text{KL}(\nu(\cdot \mid x) \parallel \mu(\cdot \mid x))\}. \quad (5)$$

The opponent ν searches for responses that perform well against π under objective k . The KL term restricts this search to a neighborhood of μ . Thus, a large β_k keeps the opponent close to μ , whereas a small β_k gives more weight to worst-case comparisons. The limit $\beta_k \downarrow 0$ is the unregularized worst-case game value $\min_{\nu} g_k(\pi, \nu)$.

Let

$$r_{k, \pi}(z \mid x) = \mathbb{E}_{y \sim \pi(\cdot \mid x)} A_k(y, z \mid x). \quad (6)$$

The minimizing opponent and the game value are

$$\nu_{k, \pi}^*(z \mid x) = \frac{\mu(z \mid x) \exp\{-r_{k, \pi}(z \mid x)/\beta_k\}}{\mathbb{E}_{z' \sim \mu(\cdot \mid x)} \exp\{-r_{k, \pi}(z' \mid x)/\beta_k\}}, \quad (7)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \rho} \log \mathbb{E}_{z \sim \mu(\cdot \mid x)} \exp\{-r_{k, \pi}(z \mid x)/\beta_k\}. \quad (8)$$

Proposition 1 (Structure). *For every k and $\beta_k \geq 0$, V_{k, β_k} is continuous and concave in π . For $\beta_k > 0$ it is differentiable, with*

$$q_{k, \pi}(y \mid x) = \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot \mid x)} A_k(y, z \mid x), \quad DV_{k, \beta_k}(\pi)[h] = \langle q_{k, \pi}, h \rangle_\rho, \quad (9)$$

where D is the directional derivative operator. The adaptive opponent makes $V_{k, \beta_k}(\pi)$ depend on the full pairwise comparison matrix, rather than only on comparisons with one fixed reference response.

Why a fixed reference can miss cyclic structure. Consider one prompt and one objective. Let δ_y denote the deterministic policy that always outputs response y , and let the reference be $\mu = \delta_r$. Suppose A , B , and C each beat r with probability 0.75, but form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. The pure policy δ_A and the uniform mixture $\pi_{\text{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have the same fixed-reference score, $P(\pi \succ r) - 1/2 = 0.25$. Their worst-case game values differ: at $\beta = 0$, response C defeats δ_A , so $V_0(\delta_A) = -0.5$, whereas the uniform mixture ties every response in the cycle, so $V_0(\pi_{\text{mix}}) = 0$. Since $V_0(\mu) = -0.25$, their improvements over the fallback are -0.25 and 0.25 , respectively. A fixed reference therefore cannot distinguish a fully exploitable pure response from a balanced mixture once both have the same reference win rate. Appendix A.2 gives the full calculation and formalizes the limiting relation to the fixed-reference precursor.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

This section converts the objective-wise game values from Section 3 into a bargaining problem over improvements relative to the reference. We state the feasibility condition under which this problem is well defined, characterize the resulting Nash solution, and compare its population guarantees with utilitarian and maxmin aggregation. The reference defines one fallback value and one improvement (the bargaining *surplus*) per objective:

$$d_k = V_{k,\beta_k}(\mu), \quad s_k(\pi) = V_{k,\beta_k}(\pi) - d_k. \quad (10)$$

A positive surplus means that π has a higher regularized game value than the deployed reference for objective k .

Assumption 1 (Joint improvement and support). *The reference μ has full support, and some policy $\bar{\pi}$ satisfies $s_k(\bar{\pi}) > 0$ for every k .*

The assumption can fail when the reference is already optimal for one objective or when no policy jointly improves the full objective set. Rather than treating positivity as a separate preprocessing problem, we expose it directly through one auxiliary surplus variable per objective:

$$\begin{aligned} (\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \\ \text{s.t. } u_k \leq s_k(\pi), \quad k \in [K], \\ \pi \in \Pi. \end{aligned} \quad (11)$$

Because every s_k is concave, each constraint is the hypograph of a concave function, and (11) is a convex optimization problem in the standard sense of maximizing a concave objective over a convex set. It is not a linear program at finite temperature because V_{k,β_k} contains the log-sum-exp in (8). The objective is strictly increasing in every u_k , so every optimum satisfies $u_k = s_k(\pi)$; hence (11) is equivalent to maximizing

$$F(\pi) = \sum_{k=1}^K \log s_k(\pi) \quad \text{over} \quad \Pi_+ = \{\pi : s_k(\pi) > 0 \ \forall k\}. \quad (12)$$

We do not add an objective-level KL term. KL appears only in the opponent definition and as the proximal geometry used to solve the same constrained objective.

Let $V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi))$ and define the comprehensive individually rational hull

$$\mathcal{G}_\beta = \{u \in \mathbb{R}^K : d \leq u \leq V_\beta(\pi) \text{ componentwise for some } \pi\}. \quad (13)$$

The next result verifies that these policy-induced game values form a valid classical bargaining problem; this is the step that justifies applying the Nash solution and its standard outcome guarantees.

Theorem 1 (Bargaining outcome). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, comprehensive, and nondegenerate. Maximizing (11) is equivalent to applying the classical Nash solution to $(\mathcal{G}_\beta, d)$. Every optimizer strictly improves every objective over the fallback and is Pareto optimal in V_β , and all optimizers induce the same game-value vector.*

The policy itself can remain nonunique because distinct response distributions may realize the same game-value vector. Beyond these core outcome guarantees, two further properties are especially relevant when objectives use different numerical scales and when improvements must be balanced across judges.

Theorem 2 (Proportional fairness and scale covariance). *Let $s^N = s(\pi^N)$ be the unique surplus vector selected by (11). Every feasible policy $\pi \in \Pi_+$ satisfies*

$$\sum_{k=1}^K \frac{s_k(\pi) - s_k^N}{s_k^N} \leq 0. \quad (14)$$

Moreover, for arbitrary $c_k > 0$ and $b_k \in \mathbb{R}$, replacing (V_k, d_k) by $(c_k V_k + b_k, c_k d_k + b_k)$ leaves the policy optimizer set unchanged.

To clarify why these guarantees favor Nash bargaining over common aggregation alternatives, Table 1 compares their population-level properties under the same joint-positive-surplus setting. Each entry is a guarantee for every global population optimizer; positive-weight utilitarian uses fixed weights $w_k > 0$, and affine covariance refers to jointly transforming each game-value coordinate and its fallback.

Table 1: Population guarantees under joint positive surplus.

rule	strict IR	Pareto optimal	unique value	scale covariance
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

Appendix A.5 gives explicit finite-game counterexamples for the other rules. Appendix A.4 also distinguishes covariance of the utility representation from rescaling raw preference labels at fixed temperature.

5 CONSTRAINED NASH BARGAINING PREFERENCE OPTIMIZATION

This section solves (11) without a separate feasibility objective. The key is an implicit entropy-proximal step whose constraints enforce positive surplus and whose dual variables recover the Nash inverse-surplus weights.

5.1 CONSTRAINED ENTROPY-PROXIMAL UPDATE

For an interior policy, Proposition 1 gives

$$\nabla F(\pi) = \sum_{k=1}^K \frac{q_{k,\pi}}{s_k(\pi)}, \quad q_{k,\pi}(y | x) = \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} A_k(y, z | x). \quad (15)$$

This explicit gradient is undefined at the reference because $s_k(\mu) = 0$. We therefore do not evaluate it there. Let $D(\pi \| \pi_t) = \mathbb{E}_x \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. Starting from $\pi_0 = \mu$, stage t solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\eta_t > 0$ is an algorithmic step size. The KL term is proximal, not part of the bargaining objective. Under Assumption 1, (16) is feasible for every center π_t , including μ , and its optimum has $u_{t+1,k} > 0$. Thus the first update itself enters the positive-surplus domain.

The Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \frac{1}{\eta_t} D(\pi \| \pi_t), \quad \lambda \geq 0. \quad (17)$$

Maximizing over $u_k > 0$ gives $u_k = 1/\lambda_k$ and the K -dimensional dual

$$\phi_t(\lambda) = -\sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad \lambda \in \mathbb{R}_{++}^K, \quad (18)$$

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t(\lambda)) - \frac{1}{\lambda_k}, \quad (19)$$

where $\pi_t(\lambda)$ is the unique maximizer in (18).

Proposition 2 (Endogenous inverse-surplus weights). *Under Assumption 1, strong duality holds for (16). At its solution,*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Hence the inverse-surplus weights are KKT multipliers of the positive-surplus constraints rather than clipped weights evaluated before the update.

5.2 SAMPLED DUAL SOLVE AND PARTITION-FREE REGRESSION

For fixed λ , the policy maximizer in (18) is characterized by an implicit pairwise log-ratio equation. For a full-support proposal over response pairs, draw y, y' and, for each objective, draw $z_k \sim \nu_{k,\bar{\pi}}^*(\cdot | x)$ for the current inner candidate $\bar{\pi}$. Query independent binary outcomes and define

$$Z_{t,k} = B_k(y, z_k) - B_k(y', z_k). \quad (21)$$

Then $\mathbb{E}[Z_{t,k} | y, y', z_k] = A_k(y, z_k) - A_k(y', z_k)$. Let

$$h_t(\pi; y, y') = \log \frac{\pi(y | x)}{\pi_t(y | x)} - \log \frac{\pi(y' | x)}{\pi_t(y' | x)}. \quad (22)$$

At the fixed point $\bar{\pi} = \pi_t(\lambda)$, the population loss

$$\mathcal{J}_{t,\lambda}(\pi) = \mathbb{E} \left[h_t(\pi; y, y') - \eta_t \sum_{k=1}^K \lambda_k Z_{t,k} \right]^2 \quad (23)$$

has $\pi_t(\lambda)$ as its unique full-support minimizer. In practice we alternate opponent reweighting and regression until the fixed-point residual stabilizes. For numerical conditioning one may use $\lambda_k = \lambda_k / \sum_j \lambda_j$ and multiply the regression coefficient by $\sum_j \lambda_j$; the target in (23) is then unchanged exactly.

The dual is only K -dimensional. Given a training estimate $\hat{s}(\pi_t(\lambda^{(m)}))$, one projected dual step is

$$\lambda^{(m+1)} = \Pi_{\Lambda} \left[\lambda^{(m)} - \gamma_m \left(\hat{s}(\pi_t(\lambda^{(m)})) - (\lambda^{(m)})^{-1} \right) \right], \quad (24)$$

where the inverse is elementwise and $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ is a broad numerical box. If the box contains the KKT multipliers, projection does not change the solution; a boundary hit triggers expansion rather than silent clipping. With exact weighted-policy solves, ϕ_t is strongly convex and smooth on Λ , so projected dual descent converges geometrically for a sufficiently small fixed γ_m ; Appendix A.6 gives the argument.

Estimating a surplus requires sampling because (8) contains expectations over prompts, learner responses, and reference comparators. Training uses a consistent plug-in estimate from sampled prompts, M reference responses per prompt, and L learner comparisons per reference response; its finite-sample bias is treated as part of the inexact-gradient error. Acceptance uses an independent or preallocated sealed batch and the simultaneous nested intervals in Appendix B. The valid surplus interval is

$$\underline{s}_k(\pi) = \underline{V}_k(\pi) - \bar{V}_k(\mu), \quad \bar{s}_k(\pi) = \bar{V}_k(\pi) - \underline{V}_k(\mu). \quad (25)$$

For any already accepted current policy, a valid lower bound on the welfare change is

$$\underline{\Delta F}_t = \sum_k \log \underline{s}_k(\tilde{\pi}_{t+1}) - \sum_k \log \bar{s}_k(\pi_t). \quad (26)$$

A candidate is accepted only if all lower surpluses are positive and, once an accepted current policy exists, $\underline{\Delta F}_t \geq 0$.

Algorithm 1 Constrained Nash Bargaining Preference Optimization

Require: Reference μ , preference oracles $P_{1:K}$, temperatures $\beta_{1:K}$, proximal steps $\eta_{0:T-1}$, dual steps $\gamma_{0:M-1}$, confidence δ .

- 1: Estimate $d_k = V_{k,\beta_k}(\mu)$ for training and cache independent simultaneous reference intervals for certification.
- 2: Set $\pi_0 \leftarrow \mu$, initialize positive dual weights, and set `accepted` $\leftarrow$ `false`.
- 3: **for** $t = 0, \dots, T - 1$ **do**
- 4: Initialize $\lambda^{(0)}$ from the previous stage (or $\mathbf{1}$ at $t = 0$).
- 5: **for** $m = 0, \dots, M - 1$ **do**
- 6: Approximate $\pi_t(\lambda^{(m)})$ by alternating regularized-opponent reweighting and regression with (23).
- 7: Estimate training surpluses $\hat{s}_k(\pi_t(\lambda^{(m)}))$ by nested sampling.
- 8: Update $\lambda^{(m+1)}$ by (24).
- 9: **end for**
- 10: Set the fitted candidate $\tilde{\pi}_{t+1} \leftarrow \pi_t(\lambda^{(M)})$ and certify it on the batch assigned to this check.
- 11: **if** some $\underline{s}_k(\tilde{\pi}_{t+1}) \leq 0$ **then**
- 12: Refine the inner solve, increase the assigned sample budget, or set $\pi_{t+1} \leftarrow \pi_t$.
- 13: **else if** `accepted` = `true` and $\underline{\Delta F}_t < 0$ **then**
- 14: Refine the inner solve or set $\pi_{t+1} \leftarrow \pi_t$.
- 15: **else**
- 16: Accept $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$ and set `accepted` $\leftarrow$ `true`.
- 17: **end if**
- 18: **end for**
- 19: **if** `accepted` = `false` **then**
- 20: **return** “no certified joint game-value improvement under the allocated budget.”
- 21: **end if**
- 22: **return** the last accepted policy.

5.3 LAST-ITERATE CONVERGENCE FROM THE REFERENCE

Theorem 3 (Exact proximal last-iterate convergence). *Under Assumption 1, let π^* maximize (11), write $s^* = s(\pi^*)$, set $\pi_0 = \mu$, and let (16) be solved exactly with arbitrary steps $\eta_t > 0$. Then π_t has full support and belongs to Π_+ for every $t \geq 1$, $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and, for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \parallel \mu)}{\sum_{t=0}^{T-1} \eta_t}. \quad (27)$$

The optimal surplus vector s^ is unique, and*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \parallel \mu)}{\sum_{t=0}^{T-1} \eta_t}. \quad (28)$$

In particular, a constant $\eta_t \equiv \eta$ gives $O(1/T)$ last-iterate welfare convergence and $O(T^{-1/2})$ surplus-vector convergence, with no upper bound on η and no positive-surplus initialization.

Appendix A.7 proves the result and gives an inexact extension. Indexing only accepted fitted stages, if their variational residuals are δ_t , their uniform game-gradient errors are ξ_t , and certification makes welfare nondecreasing, the numerator of (27) increases by $\sum_t (\delta_t + 2\eta_t \xi_t)$. Rejected proposals leave the current policy unchanged and are not counted as proximal progress.

6 EXPERIMENTS

Setup. We evaluate four objectives—helpfulness, honesty, instruction following, and truthfulness—on ULTRA FEEDBACK. The base and reference policy are Llama-3.1-8B-Instruct. A prompted Qwen3-32B preference oracle supplies objective-specific training comparisons; each pair is queried in both presentation orders and swap-averaged. Final evaluation uses Llama-3.3-70B. Appendix H gives the rubrics and label construction.

Table 2: LLM as a judge results on two backbones. All entries are swap-averaged win rates against that backbone’s own reference policy on the benchmark’s own prompts, judged by a single pinned open-weights judge, Llama-3.3-70B-Instruct, which supplied none of the training preferences. These are open-judge pairwise proxies rather than canonical GPT scores. “Alpaca LC” is the length-controlled AlpacaEval variant, “len.-matched” the Arena-Hard win rate restricted to response pairs within a $1.3\times$ length band, and “mean tokens” the mean generated Arena-Hard length; the last two are included because open judges reward verbosity, and across these arms the win rate correlates 0.58 with response length. Within each block every arm starts at $\pi_0 = \mu$, shares the pair set, optimizer and budget, and differs only in its aggregation rule. NBPO is reported at the setting its own Nash product selects and MOPO at $\tau=0.1$; the paragraph below explains both choices, how to read the margins, and why the Qwen3-8B block required suppressing that model’s reasoning trace.

method	Arena-Hard v2	AlpacaEval 2	Alpaca LC	MT-Bench	len.-matched	mean tokens
<i>Backbone and reference policy: Llama-3.1-8B-Instruct</i>						
Base = $\mu = \pi_0$	0.500	0.500	0.500	0.500	0.500	2859
NBPO (ours)	0.558	0.554	0.554	0.553	0.561	3034
NBPO, 300-step budget	0.546	0.549	0.530	0.538	0.556	2904
PROSPER/MaxEntBW	0.515	0.546	0.541	0.560	0.526	3124
Utilitarian	0.522	0.544	0.548	0.516	0.529	3045
Absolute maxmin	0.532	0.536	0.548	0.525	0.529	3130
Surplus maxmin	0.505	0.532	0.524	0.503	0.536	2978
Uniform-INPO	0.527	0.517	0.535	0.503	0.535	2976
MOPO	0.508	0.512	0.516	0.513	0.495	2862
HT-MNPO (honesty)	0.549	0.528	0.522	0.519	0.549	3006
HT-MNPO (truthfulness)	0.536	0.531	0.517	0.525	0.533	2951
HT-MNPO (helpfulness)	0.530	0.550	0.546	0.494	0.555	3127
HT-MNPO (instr. following)	0.515	0.529	0.533	0.541	0.524	2957
<i>Backbone and reference policy: Qwen3-8B, reasoning trace suppressed</i>						
Base = $\mu = \pi_0$	0.500	0.500	0.500	0.500	0.500	3215
NBPO (ours)	0.512	0.562	0.534	0.529	0.508	3402
PROSPER/MaxEntBW	0.505	0.535	0.519	0.497	0.504	3316
Utilitarian	0.509	0.528	0.511	0.497	0.507	3307
Absolute maxmin	0.510	0.531	0.514	0.503	0.503	3316
Surplus maxmin	0.500	0.534	0.523	0.516	0.505	3329
Uniform-INPO	0.491	0.498	0.496	0.481	0.490	3194
MOPO	0.498	0.499	0.497	0.468	0.493	3218
HT-MNPO (honesty)	0.493	0.500	0.503	0.513	0.494	3210
HT-MNPO (truthfulness)	0.503	0.517	0.508	0.528	0.503	3257
HT-MNPO (helpfulness)	0.491	0.497	0.498	0.519	0.497	3215
HT-MNPO (instr. following)	0.491	0.491	0.490	0.509	0.495	3205

Results.

Objective-wise surplus across datasets. Table 3 reports one panel. Table 4 repeats the measurement on two further datasets so that the aggregation rules are compared where their objectives conflict differently. The reading is mixed and we report it as such. On ULTRAFEEDBACK, NBPO is the only rule with positive surplus on every objective and it attains the largest minimum. On TL;DR it attains the largest average surplus and keeps conciseness at the fallback while the utilitarian rule and PROSPER both drive that objective clearly negative, but surplus maxmin ends with the larger minimum. On the Qwen3-8B backbone, which carries the full eleven-arm baseline set, NBPO leads all four objectives, the minimum and the average; its minimum is 55% above the next best and its average more than double. Six of the eleven arms lose individual rationality there, including PROSPER, MOPO, uniform-INPO and three of the four single-objective corners. This panel is rebuilt with Qwen3’s reasoning trace suppressed; an earlier version left it on, and 71% of that pool’s responses ended inside the trace, which flattened every arm into a 0.009 band and let a single-objective corner appear to win. The dual is nearly uniform here either way (λ between 8.2 and 10.3, against 14.5–18.0 on the LLAMA panel), so the objectives genuinely do leave similar room on this backbone; what the truncation removed was the separation between the trained policies, not between the objectives. On SAFERLHF NBPO places last of the four on both the minimum and the average, and does so even though it alone was tuned: its proximal step size was selected from a four-point sweep by held-out minimum surplus, PROSPER received the same sweep, and the two remaining rules were run at a single setting. The panels differ in how much their objectives disagree—the share of response pairs on which every objective agrees is 0.25 on TL;DR, 0.42 on ULTRAFEEDBACK and 0.50 on SAFERLHF—and SAFERLHF carries only two objectives, so it is the panel with the least for a

Table 3: Objective-wise game-value improvement over the reference on ULTRAFEDBACK, on the population scale of (2). Held-out surpluses on one 657-prompt jointly complete panel judged in a single pass by the pinned Qwen3-32B oracle; the panel spans all twelve policies, so it is smaller than the ten-policy panel of the previous draft and every row shifts by at most 0.005. “IR” is the fraction of held-out prompts on which all four objectives exceed the fallback; the base policy scores 0.254 on this statistic, so that value is the sampling floor rather than zero. “ref. win” is the mean anchored win rate. Every arm starts at $\pi_0 = \mu$, shares the pair set, optimizer and 300-step budget, and differs only in its aggregation rule; NBPO uses the proximal step size selected in Appendix G and MOPO the temperature of its own sweep that attains positive surplus. Four arms attain positive surplus on all four objectives—NBPO, MOPO, and the honesty and instruction-following corners—but NBPO’s minimum is $1.8\times$ the next largest, and it leads every column of the table.

method	instr. s	truth. s	honest s	helpful s	min s	IR	ref. win
Base = $\mu = \pi_0$	0.0000	0.0000	0.0000	0.0000	0.0000	0.254	0.493
NBPO (ours)	0.0391	0.0485	0.0546	0.1093	0.0391	0.342	0.555
PROSPER/MaxEntBW	−0.0349	−0.0508	−0.0458	0.0676	−0.0508	0.240	0.469
Utilitarian	−0.0639	−0.0765	−0.0705	0.0268	−0.0765	0.228	0.433
Absolute maxmin	−0.0434	−0.0624	−0.0512	0.0594	−0.0624	0.263	0.455
Surplus maxmin	−0.0137	−0.0004	0.0200	0.1081	−0.0137	0.295	0.516
Uniform-INPO	−0.0082	0.0222	0.0203	0.0789	−0.0082	0.300	0.517
MOPO	0.0307	0.0265	0.0215	0.0321	0.0215	0.312	0.520
HT-MNPO (truthfulness)	−0.0501	−0.0486	−0.0415	0.0231	−0.0501	0.237	0.454
HT-MNPO (helpfulness)	−0.0571	−0.0728	−0.0668	0.0500	−0.0728	0.219	0.445
HT-MNPO (honesty)	0.0215	0.0273	0.0250	0.0749	0.0215	0.304	0.529
HT-MNPO (instr. following)	0.0080	0.0257	0.0321	0.0653	0.0080	0.323	0.526

bargaining rule to arbitrate. We regard that as a scope condition on the method rather than as support for it, and we do not claim uniform dominance on held-out surplus. The pattern across the four panels is that NBPO takes the largest average surplus on three of them and the largest minimum on one, while surplus maxmin takes the largest minimum on three. That is the division the theory predicts rather than a shortfall: maximising the minimum is what surplus maxmin optimises by construction, whereas the bargaining solution trades a little of the minimum for total surplus and is the only rule among these that is invariant to how each objective’s utility is scaled.

Independence from objective utility scales. The one comparison that does not depend on which panel is chosen is the rescaling test of Theorem 2. Each objective’s utility is a von Neumann–Morgenstern payoff, fixed only up to its own positive affine transformation, so a rule whose output moves when one objective is rescaled is reading a unit that carries no preference information. Table 5 multiplies each objective’s payoff and temperature by its own factor on the frozen pool—a transformation that leaves every underlying preference unchanged—and recomputes each rule’s selection without retraining, decoding or relabelling. NBPO moves 0.01% of pairs, which is the numerical floor of the solver rather than a real deviation, while the utilitarian rule moves 5.9%, PROSPER 6.9% and the two maxmin rules 9.9% and 10.9%. The mechanism is visible in the multipliers: under $\lambda^{(A)}$ the dual moves from (16.6, 14.5, 15.9, 18.0) to (82.7, 24.1, 15.9, 10.6), absorbing the rescaling exactly. This is the empirical counterpart of Theorem 2 and, unlike the surplus comparisons above, it holds by construction rather than by dataset.

Reading the benchmark table. NBPO is reported at the setting its own objective selects, and we state that choice explicitly because it changed. A step-size sweep at a 300-step budget and a budget sweep at fixed step size produce two settings that are indistinguishable on held-out *minimum* surplus (+0.0118 at 20 η /300 steps against +0.0116 at 8 η /600 steps). The minimum is the egalitarian criterion that surplus maxmin optimises by construction; what NBPO optimises is the Nash product $\sum_k \log u_k$, and on that quantity the 600-step setting is the better iterate (−15.14 against −15.49), with a 13% larger average surplus and all four objectives still positive. The tie is therefore broken by the method’s own objective, computed on held-out game values and not on any benchmark, and the 300-step arm is kept in the table so the reader can see both. The four HT-MNPO rows are the single-objective corners, one per ULTRAFEDBACK objective.

Table 4: Objective-wise held-out surplus on three datasets, population scale of (2), with the same Llama-3.1-8B-Instruct reference and the same pinned Qwen3-32B oracle throughout. Within each panel every arm starts at $\pi_0 = \mu$, trains on the identical response pool with the identical optimizer budget, and differs only in the weight vector over objectives, so the comparison isolates the aggregation rule. Across panels the magnitudes are not comparable: ULTRAFEEDBACK is scored on the 657-prompt jointly complete panel of Table 3, while TL;DR and SAFERLHF are scored on 100 held-out prompts against four anchor samples. Objective columns follow each panel’s own header. On SAFERLHF both NBPO and PROSPER are reported at the proximal step size selected from a common four-point sweep by held-out minimum surplus, a criterion fixed before the sweep was run; the utilitarian and surplus-maxmin arms were run at a single step size and were not tuned. The fourth panel repeats the ULTRAFEEDBACK objectives on a Qwen3-8B backbone—instruction-tuned; Qwen3 unifies thinking and non-thinking modes in one checkpoint and drops the -Instruct suffix used in Qwen2.5—so that the aggregation comparison does not rest on a single base model. Its objective columns run helpfulness, truthfulness, honesty, instruction following, the order its pool was built in. Best minimum surplus in each panel is bold.

method	objective-wise surplus s_k				min s	avg s
ULTRAFEEDBACK: instruction following, truthfulness, honesty, helpfulness						
NBPO (ours)	0.039	0.049	0.055	0.109	0.039	0.063
PROSPER/MaxEntBW	-0.035	-0.051	-0.046	0.068	-0.051	-0.016
Utilitarian	-0.064	-0.077	-0.071	0.027	-0.077	-0.046
Surplus maxmin	-0.014	-0.000	0.020	0.108	-0.014	0.028
TL;DR: coverage, faithfulness, conciseness, helpfulness						
NBPO (ours)	0.198	0.079	-0.001	0.178	-0.001	0.114
PROSPER/MaxEntBW	0.205	0.031	-0.068	0.175	-0.068	0.086
Utilitarian	0.221	0.072	-0.046	0.188	-0.046	0.109
Surplus maxmin	0.081	0.090	0.018	0.066	0.018	0.064
ULTRAFEEDBACK on Qwen3-8B, trace suppressed: helpfulness, truthfulness, honesty, instr. following						
NBPO (ours)	0.072	0.018	0.043	0.031	0.018	0.041
PROSPER/MaxEntBW	0.039	-0.002	0.016	0.017	-0.002	0.018
Utilitarian	0.038	0.012	0.016	0.012	0.012	0.019
Absolute maxmin	0.041	0.005	0.018	0.012	0.005	0.019
Surplus maxmin	0.038	0.004	0.014	0.009	0.004	0.016
Uniform-INPO	-0.005	-0.001	-0.001	-0.003	-0.005	-0.002
MOPO	-0.002	-0.001	-0.005	0.001	-0.005	-0.001
HT-MNPO (helpfulness)	-0.002	-0.001	-0.001	0.002	-0.002	-0.001
HT-MNPO (truthfulness)	0.017	0.001	0.016	0.014	0.001	0.012
HT-MNPO (honesty)	0.000	0.005	0.001	-0.003	-0.003	0.001
HT-MNPO (instr. following)	-0.004	0.004	0.001	0.002	-0.004	0.001
SAFERLHF: helpfulness, harmlessness						
NBPO (ours)	0.118	0.101	—	—	0.101	0.110
PROSPER/MaxEntBW	0.145	0.116	—	—	0.116	0.131
Utilitarian	0.157	0.138	—	—	0.138	0.148
Surplus maxmin	0.150	0.143	—	—	0.143	0.147

Margins in this table are small relative to seed variation. Replicating NBPO and PROSPER at three seeds gives an Arena-Hard standard deviation of 0.014 per arm, which exceeds most of the gaps here: on the Qwen3-8B block NBPO leads all five columns, but only AlpacaEval by a margin worth naming (0.562 against 0.535), while the Arena-Hard, MT-Bench and length-matched gaps are 0.002, 0.0004 and 0.002. NBPO also produces the longest Arena-Hard responses in that block (3402 tokens against 3194–3329), which is why the length-matched column, where its lead nearly vanishes, is the one to read. Every arm on that backbone sits within 0.03 of its reference on Arena-Hard, so the block separates the rules far less than the objective-wise surpluses of Table 4, where six of eleven arms fail individual rationality outright. We therefore report this table as a check that no arm regresses in general capability, not as evidence for an aggregation rule; the surpluses of Tables 3 and 4 are what the rule is designed to move.

Table 5: Independent objective rescaling on the frozen 7,000-prompt ULTRAFEEDBACK pool. Each entry is the percentage of response pairs whose preferred element flips when objective k ’s payoff and temperature are both multiplied by λ_k , a transformation that leaves every underlying preference unchanged. Lower is better; 0 is the value implied by Theorem 2. No training, decoding, or judging is repeated. For the constrained solver the dual is re-solved from $\lambda = 1$ on the rescaled pool, exactly as a real run would: under $\lambda_k^{(A)} = (0.2, 0.6, 1.0, 1.7)$ its multipliers move from $(16.6, 14.5, 15.9, 18.0)$ to $(82.7, 24.1, 15.9, 10.6)$, absorbing the rescaling so that the weighted score is left unchanged and 0.01% of pairs flip. Unlike the panel rule it needs no floor on $1/s_k$, because λ is a strictly positive KKT multiplier by construction; the panel rule attains exact invariance here only because its floor does not bind at these rescalings.

selection rule	$\lambda^{(A)}$	$\lambda^{(B)}$	$\lambda^{(C)}$	mean
Constrained NBPO (ours)	0.01	0.01	0.00	0.01
NBPO, panel rule	0.00	0.00	0.00	0.00
NBPO, prompt-wise	2.53	2.97	2.32	2.61
Utilitarian	5.66	6.31	5.85	5.94
PROSPER/MaxEntBW	7.35	6.63	6.71	6.90
Surplus maxmin	16.02	13.57	0.00	9.86
Absolute maxmin	16.32	0.00	16.32	10.88

Suppressing the reasoning trace on the Qwen3-8B backbone. Qwen3 signals direct answering through an empty `<think></think>` block in its chat template, while the template used for training must additionally make the generation prompt a strict prefix of the rendered conversation so the assistant span can be located. Emitting the empty block unconditionally satisfies both requirements. Omitting it—which is what a plain ChatML template does, and what an earlier version of these runs used—silently puts the model into reasoning mode, and against a 1024-token decoding budget 59–76% of generations then terminate inside the trace. The oracle and the benchmark judge are then comparing unfinished reasoning rather than answers, which flattens the arms: under that template 71% of pool responses ended mid-trace, every rule attained individual rationality inside a 0.009 band, and the arm that had merely learned to reason more briefly scored best because it reached an answer most often. Every Qwen3-8B number reported here comes from the corrected template, under which no pool response and no benchmark generation is truncated mid-trace. The lower block of Table 2 gives that backbone’s benchmark comparison and Table 4 its objective-wise surpluses.

7 RELATED WORK

General pairwise and multi-objective preferences. NLHF and its scalable variants optimize directly against general pairwise preference oracles rather than fitting a scalar reward (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy regression update and an inner worst-case choice over objective weights and opponent policies (Zhang et al., 2026). NBPO instead retains one regularized game value per objective and applies reference-relative Nash bargaining across those values. The contribution is this separation of within-objective preference representation from across-objective aggregation, together with the corresponding constrained optimization, certification, and convergence analysis.

Multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize a weakest-objective rule, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These approaches provide useful control when weights, priorities, or thresholds are available. NBPO addresses a different setting: one compromise policy for a fixed objective panel without prespecified priorities. RACO applies clipped conflict-averse gradient combination to objective-specific preference losses and traces a Pareto frontier for user-specified weights (Chen et al., 2026). It is therefore a relevant frontier method, but does not select a single priority-free compromise.

Social choice and bargaining. Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards, and separately considers reward-free von Neumann solutions (Zhong et al., 2024a). Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our setting applies the Nash rule to improvements in objective-wise game values derived directly from separate pairwise preference oracles.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective alignment: an objective-wise preference game summarizes potentially cyclic pairwise feedback, and Nash bargaining selects one compromise across the resulting game values. Under joint improvement, the selected outcome is strictly above the reference in every objective, Pareto optimal, unique in game-value space, proportionally fair, and covariant to independent positive affine utility transformations. The constrained entropy-proximal formulation starts directly from the reference, and its KKT multipliers yield the inverse-surplus weights used by regularized-opponent sampling and pairwise regression. Theorem 3 gives last-iterate convergence without requiring a separately constructed interior policy, while Appendix A.2 records the fixed-reference precursor as a limiting case rather than making it a main contribution.

The guarantees apply to the selected game-value coordinates, not to every prompt, every pairwise comparison, or an external safety metric. The objective panel, reference policy, and opponent temperature remain modeling choices, and joint positive improvement can be infeasible. Worst-opponent evaluation can favor conservative mixtures and increases sampling cost. Finally, the current completed comparison contains one matched training run per method; the observed ordering must be replicated across independent seeds before it can support a performance claim.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 specifies the constrained training procedure. Appendices A–B provide the proximal-dual derivation, convergence proof, finite-sample certificate, and approximate-update analysis. Appendix D gives detailed results and the matched evaluation protocol. A submission package should include anonymized code, exact prompts, random seeds, model checkpoints, opponent pools, dual residuals, and per-stage confidence intervals.

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A PROOFS AND ADDITIONAL THEORY

Throughout the proofs, response sets are finite. The contextual case uses the ρ -weighted product of policy simplices and the ρ -weighted KL divergence. We therefore write most curvature calculations for one prompt; the contextual extension is block diagonal.

A.1 PROOF OF PROPOSITION 1

For fixed ν , the objective in (5) is affine in π . A pointwise infimum of affine functions is concave. Compactness of the finite policy simplices and continuity of the payoff imply continuity.

For $\beta_k > 0$, the minimization separates over prompts. Fix x and abbreviate $r(z) = r_{k,\pi}(z | x)$. The Lagrangian for

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) r(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\mu(z)}$$

gives $\log(\nu(z)/\mu(z)) = -r(z)/\beta_k - c_x$. Normalization yields (7), and substitution gives (8). The minimizer is unique because KL is strictly convex on the support of μ . Danskin’s theorem therefore gives

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_x \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} A_k(y, z | x),$$

where $z \sim \nu_{k,\pi}^*$, which is (9).

A.2 CONNECTION TO THE FIXED-REFERENCE PRECURSOR

Define the fixed-reference improvement

$$a_k(\pi) = g_k(\pi, \mu) = P_k(\pi \succ \mu) - \frac{1}{2}. \quad (29)$$

Skew-symmetry gives $a_k(\mu) = 0$.

Full cyclic example. For one prompt, let $\mu = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. Both $\pi_{\text{pure}} = \delta_A$ and $\pi_{\text{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have fixed-reference improvement $0.75 - 0.5 = 0.25$. At temperature zero, however, C defeats π_{pure} deterministically, so $V_0(\pi_{\text{pure}}) = -0.5$. Against each of A, B, C , the mixture wins with probability $(1/2 + 0 + 1)/3 = 1/2$, and it beats r with probability 0.75, so $V_0(\pi_{\text{mix}}) = 0$. The reference loses to each cycle response with probability 0.75, giving $V_0(\mu) = 0.25 - 0.5 = -0.25$. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\mu)$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference collapses two policies with different exploitability to the same score; the objective-wise game value retains the distinction.

Theorem 4 (Fixed-reference and worst-opponent limits). *For every policy π and $\beta_k > 0$,*

$$0 \leq a_k(\pi) - V_{k,\beta_k}(\pi) \leq \frac{1}{8\beta_k}. \quad (30)$$

If $d_{k,\beta_k} = V_{k,\beta_k}(\mu)$ and $s_{k,\beta_k}(\pi) = V_{k,\beta_k}(\pi) - d_{k,\beta_k}$, then

$$\sup_{\pi} |s_{k,\beta_k}(\pi) - a_k(\pi)| \leq \frac{1}{8\beta_k}. \quad (31)$$

Consequently, $s_{k,\beta_k}(\pi) \rightarrow a_k(\pi)$ uniformly as $\beta_k \rightarrow \infty$. If μ has full support on the finite response set, then

$$V_{k,\beta_k}(\pi) \rightarrow \min_{\nu} g_k(\pi, \nu) \quad \text{as } \beta_k \downarrow 0. \quad (32)$$

Corollary 1 (Consistency with the fixed-reference precursor). *Suppose some policy satisfies $a_k(\pi) > 0$ for every objective. For any sequence of temperature vectors with $\min_k \beta_k \rightarrow \infty$, every accumulation point of finite-temperature Nash optimizers maximizes $\sum_k \log a_k(\pi)$. Their fixed-reference improvement vectors converge to the unique utility vector selected by the precursor bargaining problem.*

A large-temperature expansion also gives

$$V_{k,\beta_k}(\pi) = a_k(\pi) - \frac{1}{2\beta_k} \mathbb{E}_x \text{Var}_{z \sim \mu} [r_{k,\pi}(z | x)] + O(\beta_k^{-2}), \quad (33)$$

so finite temperature adds a correction that depends on how unevenly the current policy performs across reference responses.

Proof of Theorem 4. Fix a prompt and let $R = r_{k,\pi}(Z | x)$ for $Z \sim \mu(\cdot | x)$. Since $A_k \in [-1/2, 1/2]$, also $R \in [-1/2, 1/2]$. Write

$$V_x = -\beta_k \log \mathbb{E} e^{-R/\beta_k}.$$

Jensen's inequality gives $\log \mathbb{E} e^{-R/\beta_k} \geq -\mathbb{E} R/\beta_k$, hence $V_x \leq \mathbb{E} R$. Hoeffding's lemma gives

$$\log \mathbb{E} \exp \left\{ -\frac{R - \mathbb{E} R}{\beta_k} \right\} \leq \frac{1}{8\beta_k^2}.$$

Therefore $V_x \geq \mathbb{E} R - 1/(8\beta_k)$. Averaging over x proves (30) because $\mathbb{E}_x \mathbb{E} R = g_k(\pi, \mu) = a_k(\pi)$.

Apply the same argument at $\pi = \mu$. Since $a_k(\mu) = g_k(\mu, \mu) = 0$, there are errors $e_\pi, e_\mu \in [0, 1/(8\beta_k)]$ such that

$$V_{k,\beta_k}(\pi) = a_k(\pi) - e_\pi, \quad V_{k,\beta_k}(\mu) = -e_\mu.$$

Thus $s_{k,\beta_k}(\pi) - a_k(\pi) = e_\mu - e_\pi$, proving (31).

For the zero-temperature limit, at each prompt

$$-\beta_k \log \sum_z \mu(z | x) e^{-r(z)/\beta_k} \rightarrow \min_{z: \mu(z|x) > 0} r(z).$$

Full support makes this the minimum over all responses, which equals $\min_{\nu} g_k(\pi, \nu)$ after averaging over prompts.

Proof of Corollary 1. Let $\epsilon_n = \max_k (8\beta_{n,k})^{-1} \rightarrow 0$ and let $s_{n,k}$ denote the corresponding game surplus. Theorem 4 gives $\sup_\pi |s_{n,k}(\pi) - a_k(\pi)| \leq \epsilon_n$. Choose a policy $\bar{\pi}$ with $m = \min_k a_k(\bar{\pi}) > 0$. For all sufficiently large n , $s_{n,k}(\bar{\pi}) \geq m/2$, so an optimizer π_n satisfies

$$F_n(\pi_n) \geq K \log(m/2).$$

Every positive game surplus is at most one. Hence each coordinate obeys $\log s_{n,k}(\pi_n) \geq F_n(\pi_n)$, and therefore $s_{n,k}(\pi_n) \geq (m/2)^K$. Thus the optimizer sequence remains uniformly away from the surplus boundary. On that compact interior region, $\sum_k \log s_{n,k}$ converges uniformly to $\sum_k \log a_k$. Compactness and the standard argmax argument imply that every accumulation point of π_n maximizes the fixed-reference objective. The fixed-reference bargaining problem has a unique utility vector, so the full sequence of anchor-surplus vectors converges to it.

Finally, the cumulant expansion

$$\log \mathbb{E} e^{tR} = t\mathbb{E}R + \frac{t^2}{2} \text{Var}(R) + O(t^3)$$

with $t = -1/\beta_k$ gives (33); boundedness makes the remainder uniform on the finite simplex.

A.3 PROOF OF THEOREM 1

Continuity on the compact policy simplex bounds $V_\beta(\pi)$, hence $\mathcal{G}_\beta$ is bounded. It is closed: if $u_n \rightarrow u$ and $u_n \leq V_\beta(\pi_n)$, compactness gives a subsequence $\pi_n \rightarrow \pi$, and continuity yields $u \leq V_\beta(\pi)$. Comprehensiveness follows immediately from the definition.

For convexity, take $u^i \leq V_\beta(\pi_i)$ for $i \in \{1, 2\}$ and $\lambda \in [0, 1]$. Objective-wise concavity gives

$$\lambda u^1 + (1 - \lambda)u^2 \leq \lambda V_\beta(\pi_1) + (1 - \lambda)V_\beta(\pi_2) \leq V_\beta(\lambda\pi_1 + (1 - \lambda)\pi_2).$$

Thus the convex combination lies in $\mathcal{G}_\beta$. Assumption 1 gives a point strictly above d , so the bargaining problem is nondegenerate.

The Nash objective is coordinatewise increasing. Hence an optimum over $\mathcal{G}_\beta$ can be moved to a dominating attainable vector $V_\beta(\pi)$ without decreasing its value, proving equivalence with (11).

The log objective tends to $-\infty$ when any surplus approaches zero. Since a strictly positive feasible surplus exists, the supremum is attained at a point strictly above disagreement. This proves existence and strict individual rationality.

If a policy π' weakly improved every game value over an optimizer π^* and strictly improved one, then every surplus would weakly increase and one would strictly increase. Strict monotonicity of the logarithm would imply $F(\pi') > F(\pi^*)$, a contradiction. Hence every optimizer is Pareto optimal.

For uniqueness of the utility outcome, suppose two maximizers π_1, π_2 induce distinct vectors. Concavity gives

$$V_\beta(\tfrac{1}{2}\pi_1 + \tfrac{1}{2}\pi_2) - d \geq \tfrac{1}{2}s(\pi_1) + \tfrac{1}{2}s(\pi_2)$$

coordinatewise. Since $G(s) = \sum_k \log s_k$ is increasing and strictly concave on $\mathbb{R}_{++}^K$,

$$\begin{aligned} F(\tfrac{1}{2}\pi_1 + \tfrac{1}{2}\pi_2) &\geq G(\tfrac{1}{2}s(\pi_1) + \tfrac{1}{2}s(\pi_2)) \\ &> \tfrac{1}{2}G(s(\pi_1)) + \tfrac{1}{2}G(s(\pi_2)), \end{aligned}$$

contradicting optimality. All maximizing policies therefore induce the same game-value vector.

A.4 PROOF OF THEOREM 2 AND PAYOFF RESCALING

Changing the utility coordinates together with their fallback values is distinct from changing the raw preference payoff while keeping the opponent temperature fixed. For $\lambda_k > 0$,

$$V_{\beta_k}^{\lambda_k A_k}(\pi) = \lambda_k V_{\beta_k/\lambda_k}^{A_k}(\pi). \quad (34)$$

Thus exact covariance at the oracle level requires jointly scaling (A_k, β_k) to $(\lambda_k A_k, \lambda_k \beta_k)$, or taking the unregularized limit.

Let $\mathcal{S}_+ = \{s \in \mathbb{R}_{++}^K : d + s \in \mathcal{G}_\beta\}$. This set is convex, and s^N maximizes $G(s) = \sum_k \log s_k$ over $\mathcal{S}_+$. First-order optimality therefore gives

$$\langle \nabla G(s^N), s - s^N \rangle = \sum_k \frac{s_k - s_k^N}{s_k^N} \leq 0$$

for every $s \in \mathcal{S}_+$. Taking $s = s(\pi)$ proves (14).

Under $V'_k = c_k V_k + b_k$ and $d'_k = c_k d_k + b_k$, the transformed surplus is $s'_k = c_k s_k$. Therefore

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k,$$

a policy-independent shift. The optimizer set is unchanged.

For raw payoff scaling,

$$\begin{aligned} V_{\beta_k}^{\lambda_k A_k}(\pi) &= \min_{\nu} \{ \lambda_k g_k^A(\pi, \nu) + \beta_k \text{KL}(\nu \| \mu) \} \\ &= \lambda_k \min_{\nu} \{ g_k^A(\pi, \nu) + (\beta_k / \lambda_k) \text{KL}(\nu \| \mu) \}, \end{aligned}$$

which is (34). Setting the transformed temperature to λ_k / β_k gives $V_{\lambda_k \beta_k}^{\lambda_k A_k} = \lambda_k V_{\beta_k}^{A_k}$ and therefore exact Nash covariance.

A.5 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

Proposition 3 (Guarantees of competing rules). *There are jointly improvable finite pairwise games in which positive-weight utilitarian aggregation selects a policy below the fallback for one objective, and instances in which absolute maxmin violates fallback-relative individual rationality. Surplus maxmin is strictly individually rational under Assumption 1, but its optimizer set can contain a policy Pareto dominated by another optimizer. Every Nash optimizer has the guarantees in Theorem 1.*

A positive-weight utilitarian objective is strictly increasing in every coordinate, so every utilitarian maximizer is Pareto optimal. It need not be individually rational. Absolute maxmin can ignore objective-specific disagreement levels. Surplus maxmin is strictly individually rational under Assumption 1, because a feasible policy has positive minimum surplus, but its flat optimum can contain dominated ties. The following skew-symmetric games make all three failures explicit.

All examples use three responses and the uniform reference $\mu = (1/3, 1/3, 1/3)$. Utilities are unregularized game values $V_k(\pi) = \min_j (\pi^\top A_k)_j$.

Utilitarian failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

Then $d = (-1/3, -1/12)$. The policy $(1/4, 1/4, 1/2)$ has surplus $(1/12, 1/48) > 0$, but equal-weight utilitarian optimization has an optimizer $(0, 1/5, 4/5)$ with surplus $(7/30, -1/60)$.

Absolute-maxmin failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again $d = (-1/3, -1/12)$. The policy $(1/3, 7/15, 1/5)$ has surplus $(1/15, 1/15) > 0$. Absolute maxmin instead has an optimizer $(0, 4/5, 1/5)$ with utility $(-1/10, -1/10)$ and surplus $(7/30, -1/60)$.

A dominated surplus-maxmin tie. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

Here $d = (-1/3, -1/6)$. Objective 2 has maximal game value zero for every mixture supported on the first two responses, so no policy can have second surplus above $1/6$. Both

$$\pi_c = (1/3, 2/3, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$\pi_e = (0, 1, 0), \quad s(\pi_e) = (1/3, 1/6),$$

therefore maximize the minimum surplus at $1/6$, but π_e Pareto dominates π_c in game values.

A.6 DUAL DERIVATION AND REGRESSION FIXED POINT

Assumption 1 supplies a strictly feasible pair for (16): choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and any $0 < u_k < s_k(\bar{\pi})$. Slater’s condition therefore gives strong duality. The Lagrangian is (17). For $\lambda_k > 0$,

$$\max_{u_k > 0} \{\log u_k - \lambda_k u_k\} = -\log \lambda_k - 1, \quad u_k^* = \frac{1}{\lambda_k},$$

where the displayed optimizer is the slack variable (not an opponent policy). If any $\lambda_k = 0$, the supremum over u_k is infinite, so the dual domain is $\mathbb{R}_{++}^K$. This proves (18).

For fixed $\lambda > 0$, the term $-D(\pi \|\pi_t)/\eta_t$ is strictly concave. Hence the policy maximizer $\pi_t(\lambda)$ is unique. Because π_t has full support and the derivative of negative KL diverges inward at a zero-probability coordinate, the maximizer also has full support. Danskin’s theorem gives

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t(\lambda)) - \frac{1}{\lambda_k},$$

which is (19). KKT stationarity in u_k gives $1/u_k - \lambda_k = 0$. Since $\lambda_k > 0$, complementary slackness makes $u_k = s_k(\pi)$, proving Proposition 2.

Inner dual convergence. Fix a box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ containing a stage dual minimizer. The Hessian of $-\sum_k \log \lambda_k$ is bounded below by $m_\Lambda I$ with $m_\Lambda = \lambda_{\max}^{-2}$, while the maximized term in (18) is convex. Hence ϕ_t is m_Λ -strongly convex on Λ . For $\beta_k > 0$, the game values are smooth; the KL term makes the weighted policy maximizer unique and interior, so the implicit-function theorem and compactness of Λ give an $L_\Lambda < \infty$ such that $\nabla \phi_t$ is L_Λ -Lipschitz. Projection is nonexpansive, and therefore exact projected dual descent with a constant $0 < \gamma < 2m_\Lambda/L_\Lambda^2$ obeys

$$\|\lambda^{(m+1)} - \lambda_t^*\|_2^2 \leq (1 - 2\gamma m_\Lambda + \gamma^2 L_\Lambda^2) \|\lambda^{(m)} - \lambda_t^*\|_2^2. \quad (35)$$

Thus the exact inner dual loop converges geometrically. Approximate policy solves and sampled surplus estimates are absorbed into the residual terms in (46).

It remains to justify the regression implementation. Fix λ and abbreviate $\bar{\pi} = \pi_t(\lambda)$. First-order optimality of the weighted policy problem in (18) gives, for each prompt and every response pair in the support,

$$\log \frac{\bar{\pi}(y \mid x)}{\pi_t(y \mid x)} - \log \frac{\bar{\pi}(y' \mid x)}{\pi_t(y' \mid x)} = \eta_t \sum_k \lambda_k (q_{k,\bar{\pi}}(y \mid x) - q_{k,\bar{\pi}}(y' \mid x)). \quad (36)$$

The conditional mean of $Z_{t,k}$ in (21), with $z_k \sim \nu_{k,\bar{\pi}}^*$, is exactly the difference on the right-hand side. The squared-loss conditional-mean decomposition therefore makes (36) necessary for every population minimizer of (23). Equality of all pairwise log differences implies that the candidate log density and $\bar{\pi}$ differ only by a prompt-dependent additive constant; normalization proves uniqueness.

A.7 PROOF OF THEOREM 3

Let $h(\pi) = \mathbb{E}_x \sum_y \pi(y | x) \log \pi(y | x)$, so its Bregman divergence is $D(\pi \| \pi_t)$. Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (37)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\mu \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (37) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_\rho \quad (38)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_\rho \quad (39)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (40)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (41)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (40), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \mu). \quad (42)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (27).

For every objective, $V_k(\pi) \in [-1/2, 1/2]$ and $d_k = V_k(\mu) \leq 0$ because $\nu = \mu$ is feasible and $g_k(\mu, \mu) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. The comprehensive positive-surplus set induced by (13) is convex, and first-order optimality of its unique maximizer s^* gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (43)$$

Combining this with (27) proves (28).

Approximate sampled stages. Reindex the accepted proposals consecutively, so t below counts accepted proximal stages and a rejected proposal contributes neither a new iterate nor a step size. Let $\hat{g}_{t+1}$ be the game-gradient estimate induced by the fitted opponents and dual weights. Suppose the fitted policy satisfies, for every $\pi \in \Pi_+$,

$$\eta_t \langle \hat{g}_{t+1}, \pi - \pi_{t+1} \rangle_\rho \leq D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t) + \delta_t, \quad (44)$$

and $\|\hat{g}_{t+1} - g_{t+1}\|_\infty \leq \xi_t$. Since the contextual ℓ_1 distance between two policies is at most two, concavity and (44) imply

$$\eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_t) - D(\pi^* \| \pi_{t+1}) + \delta_t + 2\eta_t \xi_t. \quad (45)$$

If the accepted objective values are nondecreasing, as guaranteed on the simultaneous certification event when (26) is nonnegative, then

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \mu) + \sum_{t=0}^{T-1} (\delta_t + 2\eta_t \xi_t)}{\sum_{t=0}^{T-1} \eta_t}. \quad (46)$$

Thus the fitted last iterate converges whenever the cumulative inner-solve and gradient-estimation errors are $o(\sum_t \eta_t)$. The exact theorem is the special case $\delta_t = \xi_t = 0$.

B FINITE-SAMPLE CERTIFICATION OF GAME VALUES

This section supplies the certificate invoked by Algorithm 1. It is deliberately conservative but fully explicit. Tighter empirical-Bernstein or paired bounds can replace it without changing the population objective.

B.1 NESTED ESTIMATOR

Fix at most R policy checks, including the cached reference check. Policies may be chosen adaptively, but check c uses a fresh, independent certification batch, or a sealed batch preallocated to that check before training and not revealed earlier. This independence is necessary; repeatedly reusing the same holdout after observing its outcomes is not covered by the theorem below.

For check c , draw n prompts $x_i \sim \rho$. For each prompt draw M comparators $z_{ij} \sim \mu(\cdot \mid x_i)$. For every objective, comparator, and prompt, draw L independent learner responses $y_{ij\ell} \sim \pi_c(\cdot \mid x_i)$ and binary outcomes

$$B_{k,c,i,j,\ell} \sim \text{Bernoulli}(P_k(y_{ij\ell} \succ z_{ij} \mid x_i)).$$

Define

$$\hat{r}_{k,c,i,j} = \frac{1}{L} \sum_{\ell=1}^L (B_{k,c,i,j,\ell} - \frac{1}{2}), \quad (47)$$

$$\hat{q}_{k,c,i} = \frac{1}{M} \sum_{j=1}^M \exp\{-\hat{r}_{k,c,i,j}/\beta_k\}. \quad (48)$$

Let

$$\epsilon_{\text{lab}} = \sqrt{\frac{\log(6KRnM/\delta)}{2L}}, \quad (49)$$

$$a_k = e^{-1/(2\beta_k)}, \quad b_k = e^{1/(2\beta_k)}, \quad (50)$$

$$\epsilon_{\text{cmp},k} = (b_k - a_k) \sqrt{\frac{\log(6KRn/\delta)}{2M}}, \quad (51)$$

$$\epsilon_{\text{pr}} = \sqrt{\frac{\log(6KR/\delta)}{2n}}. \quad (52)$$

For each sampled prompt define

$$\ell_{k,c,i} = -\beta_k \log \min \left\{ b_k, e^{\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} + \epsilon_{\text{cmp},k} \right\}, \quad (53)$$

$$u_{k,c,i} = -\beta_k \log \max \left\{ a_k, e^{-\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} - \epsilon_{\text{cmp},k} \right\}. \quad (54)$$

Finally set

$$\underline{V}_{k,c} = \max \left\{ -\frac{1}{2}, \frac{1}{n} \sum_{i=1}^n \ell_{k,c,i} - \epsilon_{\text{pr}} \right\}, \quad (55)$$

$$\overline{V}_{k,c} = \min \left\{ \frac{1}{2}, \frac{1}{n} \sum_{i=1}^n u_{k,c,i} + \epsilon_{\text{pr}} \right\}. \quad (56)$$

Theorem 5 (Simultaneous nested game-value intervals). *With probability at least $1 - \delta$, simultaneously for every objective $k \in [K]$ and every check $c \in [R]$,*

$$\underline{V}_{k,c} \leq V_{k,\beta_k}(\pi_c) \leq \overline{V}_{k,c}. \quad (57)$$

Consequently, if one check is the reference policy, the surplus interval in (25) is simultaneous and valid. If every lower surplus is positive, then the checked policy is feasible for (11) with probability at least $1 - \delta$. For two checked policies π_a and π_b , if every $\underline{s}_k(\pi_a) > 0$ and $\sum_k \log \underline{s}_k(\pi_a) \geq \sum_k \log \overline{s}_k(\pi_b)$, then $F(\pi_a) \geq F(\pi_b)$ on the same event.

Table 6: Exact optimization on the frozen ULTRAFEEDBACK response pool ($n = 1,372$ complete prompts). Surpluses use the population scale of Eq. (10); higher is better except for the Frank–Wolfe (FW) gap. This finite-pool certificate tests the bargaining rule, not neural generalization.

selection rule	min s_k	mean s_k	$\sum_k \log s_k$	FW gap
NBPO	0.221324	0.235957	-5.783311	9.02e-09
Utilitarian	0.217866	0.236070	-5.785865	9.86e-09

B.2 PROOF OF THEOREM 5

For a fixed sampled comparator, $\hat{r}$ averages L independent variables in $[-1/2, 1/2]$. Hoeffding’s inequality (Hoeffding, 1963) and a union bound over $KRnM$ estimates give

$$|\hat{r}_{k,c,i,j} - r_{k,\pi_c}(z_{ij} | x_i)| \leq \epsilon_{\text{lab}} \quad (58)$$

for all indices with failure probability at most $\delta/3$. The statement remains valid for adaptively selected π_c because the data used at check c are independent of the history conditional on which π_c is submitted for certification.

Let

$$q_{k,c,i} = \mathbb{E}_{z \sim \mu(\cdot | x_i)} \exp\{-r_{k,\pi_c}(z | x_i)/\beta_k\}$$

and let $q_{k,c,i}^M$ be its empirical mean over the M sampled comparators using the exact r values. Each exponential lies in $[a_k, b_k]$. A second Hoeffding bound and a union bound over KRn prompt-level means give

$$|q_{k,c,i}^M - q_{k,c,i}| \leq \epsilon_{\text{cmp},k} \quad (59)$$

with failure probability at most $\delta/3$.

On the intersection of (58) and (59),

$$e^{-\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} - \epsilon_{\text{cmp},k} \leq q_{k,c,i} \leq e^{\epsilon_{\text{lab}}/\beta_k} \hat{q}_{k,c,i} + \epsilon_{\text{cmp},k}.$$

Since $q_{k,c,i} \in [a_k, b_k]$ and $-\beta_k \log q$ is decreasing, (53) and (54) bound the exact prompt-level game value.

Each exact prompt-level value lies in $[-1/2, 1/2]$. A third Hoeffding bound over the n prompts, union-bounded over KR checks, gives prompt-generalization error at most ϵ_{pr} with failure probability at most $\delta/3$. Combining the three events proves (57). The surplus statement follows by subtracting the worst compatible endpoints of the candidate and reference intervals.

C FINITE-POOL POPULATION CERTIFICATE

We freeze 1,500 ULTRAFEEDBACK prompts, four policy responses and four reference responses per prompt, four objective-specific pairwise tensors, $\beta_k = 0.25$, and the reference game values before optimization. The complete intersection contains 1,372 prompts. We optimize (11) directly over one four-simplex per prompt in float64. The reported checkpoint is the converged final iterate, not a reward-selected iterate.

The Nash surplus vector is (0.229230, 0.258926, 0.221324, 0.234350), so every coordinate is strictly above disagreement. Its normalized Frank–Wolfe gap is 9.02×10^{-9} ; the corresponding positive supporting weights certify that no policy can improve every game value by more than this residual. A deterministic feasible restart agrees in game-value space within 3.03×10^{-9} . Scaling the four payoff, temperature, and disagreement coordinates by (0.2, 0.6, 1.0, 1.7) gives a scale-corrected surplus residual of 2.78×10^{-17} . The utilitarian control attains a slightly larger mean surplus, while the Nash solution attains the larger log product and minimum surplus. These calculations verify the finite-pool selection rule, but do not close the separate neural-regression error in Theorem 3.

Solution-concept stress test. We additionally preregister 200 new three-objective, five-action cyclic games, conditioned only on the joint-improvement assumption required by Theorem 1. We solve

Table 7: Exact population solutions on 200 preregistered cyclic games. “Defined” is the fraction for which the method returned a feasible policy, IR is the fraction strictly above the common fallback in every objective, and the Pareto gap is measured in the common game-value coordinates. MOPO uses objective 1 as primary and preregistered half-attainable thresholds. Methods optimize different social objectives, so the minimum surplus is diagnostic rather than a universal ranking.

selection rule	defined	strict IR	mean min. surplus	Pareto gap
NBPO	1.000	1.000	0.0289	1.64×10^{-13}
PROSPER/MaxEntBW	1.000	0.665	0.0071	1.86×10^{-13}
Utilitarian	1.000	0.595	-0.0009	1.42×10^{-13}
Surplus maxmin	1.000	1.000	0.0361	2.16×10^{-13}
MOPO	0.510	0.035	-0.0887	2.67×10^{-2}
HT single-objective corners	1.000	0.030–0.060	-0.1351–-0.1207	$5.95\text{--}6.75 \times 10^{-13}$

every population problem to numerical convergence and compare the resulting selection rules on the same game tensors. Table 7 separates the properties of the selection rule from neural approximation.

NBPO has zero Nash-welfare regret by construction, satisfies strict IR in every feasible game, and has maximum Pareto gap 4.36×10^{-13} . Under independent coordinate scaling by (0.25, 0.60, 1.40), its maximum policy ℓ_1 drift is 7.01×10^{-7} , whereas PROSPER’s mean drift is 0.277. Surplus maxmin obtains the larger minimum surplus, as expected from its objective, while utilitarian obtains the larger mean surplus. The experiment therefore verifies the properties that distinguish the Nash rule; it is not evidence that one neural optimizer dominates every alternative metric.

D EXPERIMENTAL PROTOCOL AND DETAILED RESULTS

Run matrix. For each dataset and training run, initialize the constrained game-value aggregation rules from the same reference policy and use the same proximal-stage budget. Hold the prompt split, decoding parameters, candidate count, comparator pool, labeled pair set, total preference-query budget, total optimizer budget, and $\beta_{1:K}$ fixed within a seed. Count inner dual iterations, opponent scoring, rejected stages, certification queries, and validation used for hyperparameter selection. PROSPER, fixed-reference precursor, MOPO, and Uniform-INPO are external baselines and should retain their canonical update rules while receiving the same total resource budget.

Independent evaluation. Training labels, utility certification, and final evaluation must use disjoint data. The final game values use an independently generated comparator pool and evaluator families that did not provide training labels. Report failure to certify the first constrained iterate as a result rather than replacing it with a clipped or offset Nash objective.

Primary metrics. For a final release, report V_{k,β_k} and s_k on the theoretical scale of (2), together with the minimum surplus, average game value, Nash welfare, IR rate, fixed-reference win rate, KL to the reference, response length, opponent entropy, and accepted-stage rate. The currently archived summary provides only the implementation-scale quantities $\tilde{s}_k$ and $\tilde{F}$ shown below. Rows without a canonical implementation audit are omitted. Nash welfare is undefined when a held-out surplus is nonpositive; do not apply an unreported offset.

D.1 FROZEN REPORTING TABLES FOR THE MATCHED NEURAL STUDY

The following tables fix the reporting surface for the new matched-budget study before its final evaluation. A dash denotes a cell that has not yet been measured; it must not be imputed from a different model, judge, prompt split, or training budget. The primary panel uses four global criteria on held-out UltraFeedback prompts: instruction following, truthfulness, honesty/calibration, and helpfulness. All methods share one response pool and comparison tensor. PROSPER retains its alternating objective-weight and opponent updates, while MOPO uses instruction following as the preregistered primary objective with fixed lower bounds on the other three objectives. The four HT-MNPO rows are single-objective corner policies rather than candidate compromise policies.

Table 8: Main comparison from the available completed matched run, on the population scale of (2). “min s ” is the smallest objective improvement, $F = \sum_k \log s_k$, and “ref. win” is the average conventional win rate against the reference. Higher is better within a panel. Detailed objective values are in Appendix D. PROSPER and MOPO are omitted pending evaluation with their canonical algorithms.

method	HELPSTEER2			SAFERLHF		
	min s	F	ref. win	min s	F	ref. win
NBPO (ours)	0.104	−8.011	0.537	0.500	−1.319	0.817
Utilitarian	0.109	−7.931	0.560	0.510	−1.314	0.821
Absolute maxmin	0.129	−6.978	0.577	0.496	−1.325	0.811
Surplus maxmin	0.133	−7.286	0.575	0.497	−1.332	0.817
Fixed-reference precursor	0.134	−7.251	0.565	0.520	−1.278	0.837
Uniform-INPO	0.079	−9.156	0.514	0.495	−1.328	0.804

Table 9: Cross-domain check on TL;DR summarisation. Held-out surpluses on 100 prompts disjoint from training, population scale of (2), judged by the same pinned Qwen3-32B oracle and anchored on the same Llama-3.1-8B-Instruct reference used throughout. Every arm starts at $\pi_0 = \mu$, trains on the identical eight-response pool with the identical optimizer budget, and differs only in the weight vector over objectives, so the comparison isolates the aggregation rule. The panel is chosen because its objectives conflict by construction—coverage and conciseness cannot both be maximised—and they do: their game gradients correlate 0.203 against a mean of 0.399 across all pairs, where the ULTRAFEEDBACK panel averages 0.616. That conflict separates the rules here in a way it does not there: each rule now buys its favoured objective at a visible cost in another, and no arm is uniformly best. The SAFERLHF panel is not yet built and is omitted rather than left blank.

method	TL;DR summarisation, four objectives					
	coverage s	faithful s	concise s	helpful s	min s	avg s
Constrained NBPO (ours)	0.198	0.079	−0.001	0.178	−0.001	0.114
PROSPER/MaxEntBW	0.205	0.031	−0.068	0.175	−0.068	0.086
Utilitarian	0.221	0.072	−0.046	0.188	−0.046	0.109
Surplus maxmin	0.081	0.090	0.018	0.066	0.018	0.064

These tables serve different claims. Table 3 tests whether the learned compromise approaches the intended bargaining outcome under genuine objective conflict. Table 2 checks that gains are not caused by verbosity or broad capability regression. Table 9 tests whether the result transfers to safety and summarization rather than depending on one prompt distribution.

Table 10: Detailed finite- β HELPSTEER2 results from the available matched run, on the population scale of (2). “IR” is the number of runs with positive improvement on every objective.

method	helpful s	correct s	coherent s	concise s	min s	F	IR
NBPO	0.197	0.134	0.104	0.121	0.104	−8.011	1/1
Utilitarian	0.199	0.132	0.109	0.125	0.109	−7.931	1/1
Absolute maxmin	0.221	0.164	0.129	0.198	0.129	−6.978	1/1
Surplus maxmin	0.233	0.159	0.133	0.139	0.133	−7.286	1/1
Fixed-reference precursor	0.223	0.154	0.134	0.154	0.134	−7.251	1/1
Uniform-INPO	0.170	0.099	0.080	0.079	0.079	−9.156	1/1

Remaining reporting checks. A final multi-seed evaluation should document the payoff normalization, plot each run’s theoretical minimum surplus, Nash welfare, and dual residual across constrained stages, mark rejected steps, test opponent-pool sensitivity, audit generic or excessive-refusal behavior, and report both game values and conventional fixed-reference win rates.

Table 11: Detailed finite- β SAFERLHF results from the available matched run, on the population scale of (2). “IR” is the number of runs with positive improvement on every objective.

method	helpful s	harmless s	min s	F	IR
NBPO	0.500	0.534	0.500	-1.319	1/1
Utilitarian	0.510	0.527	0.510	-1.314	1/1
Absolute maxmin	0.496	0.536	0.496	-1.325	1/1
Surplus maxmin	0.497	0.531	0.497	-1.332	1/1
Fixed-reference precursor	0.520	0.535	0.520	-1.278	1/1
Uniform-INPO	0.495	0.535	0.495	-1.328	1/1

Table 12: Temperature and certification ablation on the population scale of (2). “Gap” is the reported difference from the fixed-reference precursor. It should not be compared numerically with Theorem 4 until the reporting-scale normalization is documented.

β	min s	F	reported gap	opp. entropy	accepted stages	ref. win
∞	0.520	-1.278	0	n/a	2/2	0.837
1.0	0.487	-1.369	-0.034	0.988	2/2	0.803
0.25	0.500	-1.319	-0.020	0.877	2/2	0.817
0.05	0.475	-1.385	-0.046	0.592	2/2	0.802
0 (finite game only)	0.569	-1.078	-0.019	0	n/a	n/a

Table 13: Raw-label scaling intervention. The coupled condition scales both A_k and β_k by λ and should preserve the population target; the fixed-temperature condition intentionally changes the comparator neighborhood.

condition	λ	target cosine	policy shift	min $\tilde{s}$	opp. entropy
coupled (A_k, β_k)	0.6	1.000000	0.195	0.600	0.877
coupled (A_k, β_k)	0.3	1.000000	0.471	0.300	0.877
fixed β_k	0.6	0.993629	0.273	0.751	0.941
fixed β_k	0.3	0.972431	0.476	0.556	0.983

E ULTRAFEDBACK-SCALE MATCHED PROJECTION STUDY

This study instantiates the matched-comparison protocol of Appendix D on 7,000 ULTRAFEDBACK prompts with four objectives (instruction following, truthfulness, honesty, helpfulness), a pinned Qwen3-32B guided-choice judge queried in both presentation orders, four candidate responses and four reference anchors per prompt (896,000 judgments per side), and a 6,000/1,000 train/held-out prompt split. All aggregation rules share one fixed parent policy, one frozen pair set, and one optimizer budget.

Target reliability across independent candidate halves. Table 14 reports the split-half reproducibility of each rule’s centered per-response target: the target is computed twice from disjoint two-candidate halves, and the two versions are correlated over all held-out responses with a paired 2,000-resample prompt bootstrap. Every objective has strictly positive policy-minus-reference surplus in each half. The NBPO target is significantly more reproducible than both maxmin targets and statistically indistinguishable in practical terms from the utilitarian target, consistent with the near-uniform inverse-surplus weights that a balanced panel induces.

Table 14: Split-half reliability of the centered selection targets on 7,000 ULTRAFEDBACK prompts. “argmax” is the fraction of prompts whose top-scored response agrees between halves. Δ is NBPO minus the row’s rule (paired bootstrap 95% interval).

selection rule	split-half Pearson [95% CI]	argmax	Δ vs. NBPO [95% CI]
NBPO	0.884 [0.879, 0.888]	0.787	—
Utilitarian	0.894 [0.890, 0.897]	0.798	−0.010 [−0.012, −0.009]
Absolute maxmin	0.869 [0.864, 0.875]	0.932	+0.014 [+0.008, +0.020]
Surplus maxmin	0.639 [0.629, 0.649]	0.617	+0.244 [+0.236, +0.253]

Held-out realization of the regression target. Table 15 measures whether the trained network reproduces its prescribed mirror target on held-out prompts: the pairwise policy log-ratio is correlated against the frozen centered target on 148 held-out prompt groups after training on 1,340 groups for exactly four passes. Same-split memorization unit tests (32 groups, 50 passes) exceed Pearson 0.98 for every configuration shown, so optimizer and capacity are not the binding constraint; the target itself has split-half reliability above 0.88, so label noise is not the ceiling either. Across the logistic and squared losses, LoRA ranks 64 and 256, and a full-precision forward pass, held-out realization plateaus near 0.36, far below the 0.80 preregistered for opening reward-based selection. We are not aware of prior preference-optimization work that reports this quantity; the plateau motivates the pair-oriented projection used in Table 3.

Table 15: Held-out realization of the prescribed regression target (ULTRAFEDBACK, 1,340 train / 148 held-out prompt groups, four passes, best learning rate per grid, fp32 forward). “KL” is the median per-prompt KL from the parent.

projection	adapter	train Pearson	held-out Pearson	held-out KL
Logistic (Bradley–Terry)	LoRA-64	0.954	0.366	0.018
Squared (23)	LoRA-64	0.953	0.366	0.018
Squared (23)	LoRA-256	0.928	0.363	0.018

Rescaling test: construction and caveats. Theorem 2 states that the NBPO solution is unchanged when each objective’s utility is rescaled independently. Table 5 tests this on the frozen pool without retraining or relabelling: each objective’s payoff and temperature are multiplied by its own factor λ_k , which leaves the underlying preferences untouched, and every rule’s selection is recomputed. We report the fraction of the $6,000 \times 6$ response pairs whose preferred element changes. NBPO does not move a single pair under any of the three rescalings, matching the theorem exactly. Every competing rule does: the utilitarian average and PROSPER’s per-prompt worst objective both compare raw values across objectives, and the two maxmin rules move most of all. The zeros in the maxmin rows are coincidental rather than structural—they occur only when a rescaling happens to leave

the arg-minimising objective unchanged—whereas the NBPO zeros hold for every rescaling. The prompt-wise NBPO variant is close to invariant but not exactly so, because a prompt whose estimated surplus is nonpositive needs a numerical floor and the floor does not commute with rescaling; the same floor is what makes the panel-level rule deviate when it binds.

The rescaling vectors, in the objective order (helpfulness, honesty, instruction following, truthfulness), are $\lambda^{(A)} = (0.2, 0.6, 1.0, 1.7)$, $\lambda^{(B)} = (3.0, 0.5, 1.0, 0.25)$ and $\lambda^{(C)} = (1.0, 1.0, 1.0, 5.0)$.

Matched game-value comparison. Table 3 in Appendix D.1 evaluates all arms on the 1,000 held-out prompts under the frozen evaluator, anchors, and decoding shared with the parent run. “pair-projection” arms orient each frozen response pair by the rule’s selector score and train with a standard length-normalized preference loss; “exact regression” arms regress the pairwise log-ratio difference onto the rule’s signed score difference with the manuscript loss (23). Both use the identical 1,200-step budget. The completed rows stress-test the certificate requirement of Algorithm 1. Run open-loop at four times the manuscript’s optimizer budget, the squared-loss update drives every rule to the same collapsed policy: mean response length grows from 1,399 to about 2,700 characters, the anchored win rate falls to 0.07–0.09, and every held-out surplus is strongly negative. Restoring the manuscript’s own 300-step budget removes the collapse—response lengths stay within $1.2\text{--}1.3\times$ the parent and generations remain fluent—and the aggregation rules then separate measurably. NBPO dominates the utilitarian rule on every objective, on the minimum surplus, and on the anchored win rate, which is the comparison the inverse-surplus weights are designed to affect; both maxmin rules place ahead of NBPO on this panel, with surplus maxmin closest to the fallback. Critically, no trained arm at either budget attains joint positive surplus, so the constrained algorithm would reject every candidate. Since the same collapse appears under every rule, it is attributable to the uncertified projection step rather than to the aggregation rule, and it is consistent with the held-out realization plateau of Table 15: when the fitted update does not track its prescribed target off-distribution, constrained stage acceptance is the mechanism that prevents drift. We therefore report these runs as a negative result about uncertified projection, not as evidence for or against the bargaining rule.

F FIXED-REFERENCE PRECURSOR EXPERIMENTS

The tables below preserve the numerical results from the preceding NBPO draft. These runs use the anchor margin $a_k(\pi) = P_k(\pi \succ \mu) - 1/2$, add $-0.05\text{KL}(\pi\|\mu)$ to the welfare objective, initialize directly at μ , and use clipped inverse-surplus weights. By Theorem 4, they concern the $\beta = \infty$ endpoint, not finite-temperature NBPO. They are retained as an audit trail and must not be cited as evidence for the regularized-opponent objective.

Population optimizer audit. The full and minimal exact-gradient updates were identical, and absorbing the sum of inverse-surplus weights into the step size changed the policy by at most $2.9e - 15$ in ℓ_1 . A stage-level acceptance test reduced regret under frozen gradient noise by 0.0372, whereas raw inverse weights at the same nominal step reduced the strict-IR rate by 4.0%. The scale-covariance and Pareto residuals were below $2.4e - 15$. These archived checks support preserving the exact coefficient scale and retaining stage acceptance, but they do not evaluate the new constrained proximal solver.

The simultaneous Hoeffding lower bound made no false certifications in this study, compared with 1.09% for the plug-in rule at $n = 128$, but its power was only 9.0% at $n = 8192$. We therefore use confidence bounds as external certificates rather than as an optimization ingredient.

Interpretation. The precursor runs provide no reliable finite- β comparison. The three-seed HELP-STEER2 ordering changes sign, and repeated fixed-reference updates can leave the positive-surplus region, especially when the effective step scale is not controlled. These observations motivate matched seed-level evaluation, constrained proximal updates, and stage-wise certification, but they are not performance claims for NBPO.

Table 16: Four-objective HELPSTEER2 fixed-reference precursor results with a prompted pairwise oracle. Entries report win rates against the reference, averaged over Phi-4 and Llama-3.3-70B, neither of which supplied training preferences. The upper block reports four-stage runs, while the lower block reports single-stage baselines for context. Results are from one training seed.

rule	helpful	correct	coherent	concise	Avg	Worst
<i>four stages each</i>						
NBPO-KL	0.660	0.642	0.555	0.471	0.582	0.471
Utilitarian	0.625	0.597	0.530	0.426	0.545	0.426
MaxMin	0.463	0.424	0.382	0.369	0.410	0.369
<i>single stage</i>						
Utilitarian, 1 stage	0.599	0.583	0.559	0.416	0.539	0.416
MaxMin, 1 stage	0.604	0.587	0.559	0.420	0.543	0.420
HT-MNPO-helpful	0.654	0.618	0.593	0.324	0.547	0.324
HT-MNPO-correct	0.629	0.617	0.574	0.381	0.550	0.381
HT-MNPO-coherent	0.631	0.596	0.579	0.351	0.539	0.351
HT-MNPO-concise	0.422	0.470	0.448	0.722	0.515	0.422
INPO	0.616	0.604	0.560	0.425	0.551	0.425
SimPO	0.618	0.610	0.560	0.429	0.554	0.429
IPO	0.609	0.589	0.553	0.424	0.544	0.424
DPO	0.603	0.574	0.561	0.417	0.539	0.417
Base (μ)	0.500	0.500	0.500	0.500	0.500	0.500

Table 17: Three training seeds of the two four-stage fixed-reference rules on HELPSTEER2. Scored by Phi-4 on 500 held-out prompts. The $\pm$ figures are standard deviations over training seeds. The ordering does not persist across seeds.

rule	seed 42	seed 43	seed 44	mean
<i>worst objective</i>				
NBPO-KL	0.504	0.398	0.328	0.410 $\pm$.088
Utilitarian	0.461	0.490	0.318	0.423 $\pm$.092
<i>average objective</i>				
NBPO-KL	0.593	0.449	0.385	0.475 $\pm$.106
Utilitarian	0.551	0.601	0.373	0.508 $\pm$.120

Table 18: SAFERLHF fixed-reference precursor results. Entries are win rates against Llama-3.1-8B-Instruct on 500 held-out prompts, averaged over Phi-4 and Llama-3.3-70B. The subscript on NBPO-KL is the spread over three training seeds.

rule	helpful	harmless	Avg	Worst
NBPO-KL	0.807	0.845	0.826	0.807 $\pm$.014
Utilitarian	0.816	0.845	0.830	0.816
MaxMin	0.806	0.853	0.830	0.806
HT-MNPO-helpful	0.803	0.784	0.794	0.784
HT-MNPO-harmless	0.769	0.853	0.811	0.769
INPO	0.808	0.816	0.812	0.808
SimPO	0.780	0.764	0.772	0.764
IPO	0.753	0.727	0.740	0.727
DPO	0.704	0.716	0.710	0.704
Base (μ)	0.500	0.500	0.500	0.500

Table 19: Fixed-reference target under symmetric coin-flip contamination of the helpfulness judge on HELPSTEER2. The exact proportionality is a property of the $\beta = \infty$ anchor limit. It is not the prediction for finite-temperature NBPO with fixed β .

rule	λ	ω helpful	ω concise	mean $ t $	cosine	ratio spread
NBPO-KL	1.0	0.076	0.470	0.393	n/a	n/a
NBPO-KL	0.6	0.120	0.447	0.374	1.000000	8×10^{-15}
NBPO-KL	0.3	0.214	0.399	0.334	1.000000	5×10^{-15}
Utilitarian	1.0	0.250	0.250	0.403	n/a	n/a
Utilitarian	0.6	0.250	0.250	0.363	0.999105	1.28
Utilitarian	0.3	0.250	0.250	0.332	0.996738	2.24
MaxMin	any	0.000	1.000	0.411	1	0

Table 20: Fixed-reference ablation of weight normalization and pair selection on SAFERLHF. Entries are reference win rates on 500 held-out prompts. The sharp degradation without normalization motivates treating the normalized coefficient as an explicit effective step size.

weights	pairs	stage	Qwen3-32B		Phi-4	
			Avg	Worst	Avg	Worst
normalized	best-of-4	1	0.737	0.722	0.772	0.747
normalized	best-of-4	2	0.698	0.684	0.724	0.710
normalized	random	1	0.730	0.699	0.745	0.707
normalized	random	2	0.667	0.635	0.665	0.608
unnormalized	best-of-4	1	0.490	0.457	0.499	0.490
unnormalized	best-of-4	2	0.095	0.045	0.032	0.028
unnormalized	random	1	0.510	0.478	0.508	0.503
unnormalized	random	2	0.380	0.349	0.363	0.355

Table 21: Preregistered population-optimizer audit over 200 finite cyclic games. Regret differences are paired means with 95% bootstrap intervals.

comparison	endpoint	result
full vs. minimal	max policy ℓ_1	$0.0e + 00$
normalized vs. absorbed step	max policy ℓ_1	$2.9e - 15$
noisy stage gate	Nash-regret change	$-0.0372 [-0.0407, -0.0340]$
raw fixed step	strict-IR change	$-0.040 [-0.070, -0.015]$

G PROXIMAL STEP-SIZE SWEEP

Theorem 3 holds for arbitrary $\eta_t > 0$: the step size enters only through $1/\sum_t \eta_t$ in the rate, not as a constraint on validity. It is therefore an algorithm parameter rather than a tuned hyperparameter, and the value used in Table 2 is the one this sweep selects. All five settings share the dual solution, the pair set, the optimizer and the 300-step budget; only the coefficient multiplying the regression target changes.

Table 22: Held-out effect of the proximal step size on the ULTRA FEEDBACK panel (711 jointly complete prompts, population scale). “ratio” is the mean Arena-Hard generation length divided by the base policy’s. Minimum surplus rises sevenfold to 20η and then falls, while the length ratio decreases monotonically to 20η ; at 50η the regression target has RMS 21.5, the scale at which the archived open-loop arms collapsed at four times their budget, yet generations remain fluent.

step size	instr. s	truth. s	honest s	helpful s	min s	ratio
η	0.0086	0.0051	0.0207	0.1037	0.0051	1.03
3η	0.0324	0.0231	0.0366	0.1189	0.0231	1.02
8η	0.0267	0.0429	0.0519	0.1113	0.0267	1.02
20η	0.0390	0.0480	0.0542	0.1096	0.0390	1.01
50η	0.0095	0.0547	0.0505	0.1024	0.0095	1.03

What happens when every rule is given twice the budget. The step-size sweep varies η at a fixed 300-step budget. Table 23 varies the budget instead, doubling it to 600 steps for NBPO and for the two strongest baselines. Each arm is run from its own configuration with `max_steps` as the only field changed, so nothing distinguishes the doubled-budget arm from the published one except the budget. The rules do not respond alike. PROSPER and the utilitarian rule both collapse: every objective falls to roughly -0.09 and individual rationality is lost outright. NBPO at 8η keeps all four objectives positive and improves its average surplus, and only loses individual rationality at the largest step size, where the total displacement $\eta \times \text{steps}$ is twice that of any setting we report. The asymmetry is not explained by displacement alone: the baselines’ targets are normalised to an RMS of 0.43 against NBPO’s 3.44, so they move the policy *less* per step and still fail first. Their selectors are winner-take-all—PROSPER keeps only each prompt’s worst objective and the utilitarian rule a single fixed average—and a longer run against such a target appears to drive the policy off the distribution the target was estimated on, whereas the bargaining weights keep every objective represented in the regression. We report the measurement and offer that reading as a hypothesis rather than a demonstration.

Table 23: Doubling the optimizer budget from 300 to 600 steps. Held-out surpluses on 100 prompts under the pinned Qwen3-32B oracle, population scale of (2). Each arm is trained from its own published configuration with `max_steps` as the only altered field; η is the proximal step size as a multiple of the base target RMS. “IR” marks whether all four objectives remain positive.

method	setting	instr.	truth.	honest	min s	IR
NBPO	20η , 300 steps	0.0185	0.0118	0.0159	0.0118	yes
NBPO	8η , 600 steps	0.0166	0.0116	0.0217	0.0116	yes
NBPO	20η , 600 steps	−0.0101	0.0003	−0.0018	−0.0101	no
PROSPER/MaxEntBW	600 steps	−0.0858	−0.0935	−0.0963	−0.0963	no
Utilitarian	600 steps	−0.0882	−0.0900	−0.0921	−0.0921	no

H JUDGE INSTANTIATION AND PREFERENCE LABELING

Training judges. Use the same prompted pairwise interface on both panels. For each objective, Qwen3-32B compares two responses under an attribute-specific rubric. Each pair is queried in both presentation orders, and the verdicts are swap-averaged. The fixed-reference precursor also contains deterministic Beaver reward and cost heads; those runs should not be conflated with the prompted-oracle main comparison.

Evaluation judges. Evaluation is separate from training and uses Phi-4 and Llama-3.3-70B. Each evaluator returns `[[A]]`, `[[B]]`, or `[[TIE]]`, scored as 1, 0, or 1/2, in both presentation orders. This makes the empirical evaluation oracle skew-symmetric after averaging.

Rubrics. Helpfulness asks which response better accomplishes the user’s stated goal; correctness asks which is more accurate; coherence asks which is clearer and better organized; conciseness asks which conveys the needed content with less padding; and harmlessness asks which is safer. The judge is instructed to evaluate only the designated objective.

Prompt template. For concreteness, the helpfulness system message and pairwise user message are

```
System: You compare two AI responses on HELPFULNESS ONLY:
which better helps the user accomplish their stated goal
(relevance, substance, actionable detail). Ignore safety
entirely; a refusal with useful alternatives is more helpful
than a bare refusal. End with exactly [[A]] or [[B]] or
[[TIE]].
User: [User prompt] {prompt} [Response A] {A} [Response B]
{B} Verdict:
```

The other objectives use the same template with only the rubric changed, and each pair is queried again after swapping responses A and B.

From verdicts to regularized opponents. For a current policy π_t and a reference comparator z , estimate

$$\hat{r}_{k,\pi_t}(z | x) = \frac{1}{L} \sum_{\ell=1}^L (B_k(y_\ell, z) - \frac{1}{2}), \quad y_\ell \sim \pi_t(\cdot | x).$$

Reference responses are reweighted by $\exp\{-\hat{r}_{k,\pi_t}(z | x)/\beta_k\}$ and normalized to sample $z_k \sim \hat{\nu}_{k,\pi_t}^*$. For a policy pair (y, y') sharing z_k , the binary difference in (21) enters the regression target directly. No scalar reward model or value function is fitted by NBPO.